



DNA Fingerprinting

Assistant Lecturer

Hazem Abbas Al-Bustany

M.Sc. in Molecular Biology

hazem.albustany@hmu.edu.krd



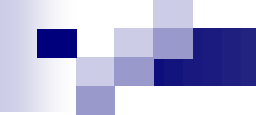
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DNA fingerprinting

Also known as **DNA profiling**, is a molecular technique used to identify individuals based on unique patterns in their DNA. With the exception of identical twins, every individual possesses a unique DNA sequence. DNA fingerprinting exploits variations in specific regions of the genome to distinguish one person from another. Since its discovery, DNA fingerprinting has become an essential tool in forensic science, medical genetics, anthropology, and legal investigations.



Historical Background

DNA fingerprinting was first developed in 1984 by **Sir Alec Jeffreys** at the University of Leicester. He discovered that certain regions of human DNA contain repetitive sequences that vary greatly among individuals. This groundbreaking discovery revolutionized forensic science and introduced DNA-based identification into criminal justice systems worldwide.



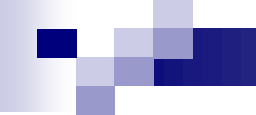
Basis of DNA Fingerprinting

Although more than **99.9%** of human DNA is identical among individuals, small variations known as **polymorphisms** account for genetic diversity. DNA fingerprinting focuses on highly variable, non-coding regions of the genome, particularly:

Variable Number Tandem Repeats (VNTRs)

Short Tandem Repeats (STRs)

These regions differ in the number of repeat units between individuals, creating a unique genetic profile.



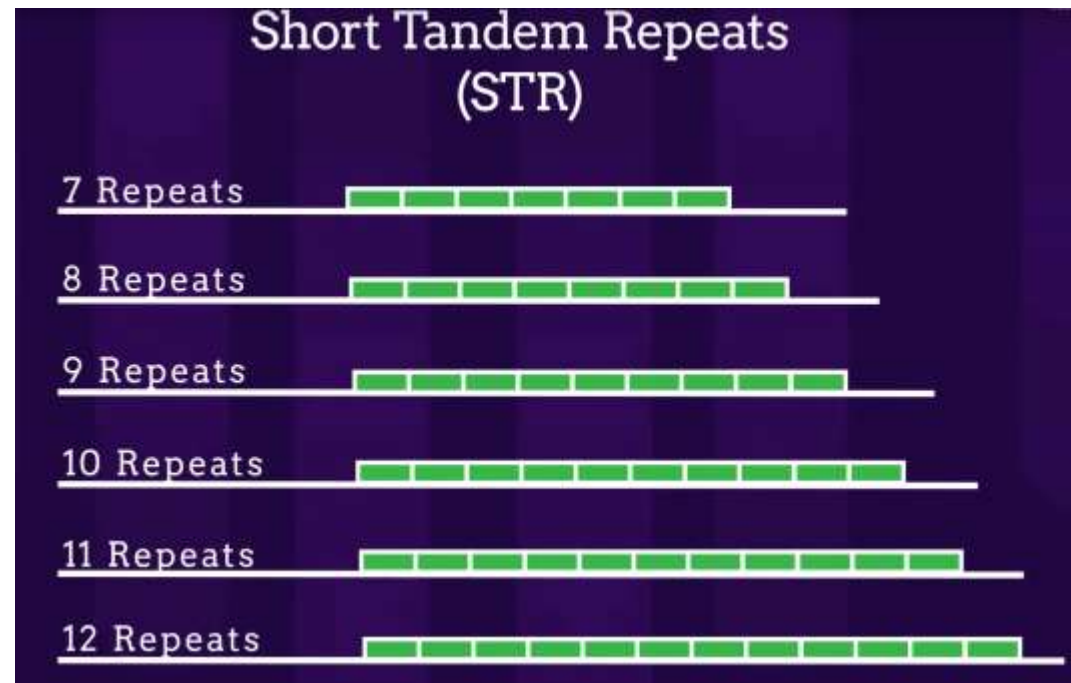
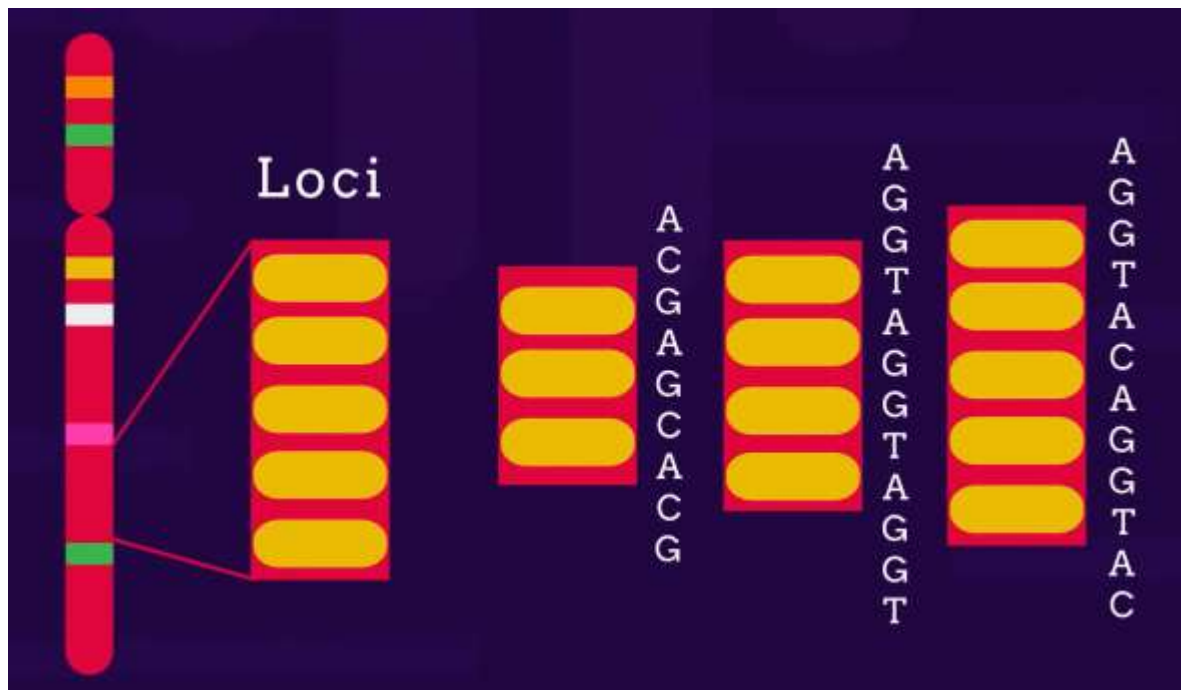
Types of DNA Polymorphisms Used

1) Variable Number Tandem Repeats (VNTRs)

VNTRs consist of long repetitive DNA sequences repeated multiple times. Early DNA fingerprinting techniques relied heavily on VNTR analysis but required large amounts of high-quality DNA.

2) Short Tandem Repeats (STRs)

STRs are short DNA sequences (2–6 base pairs) repeated in tandem. STR analysis is currently the gold standard due to its high sensitivity, reproducibility, and suitability for degraded samples.





Steps in DNA Fingerprinting

The standard procedure for DNA fingerprinting includes the following steps:

1) Sample Collection

DNA can be obtained from blood, saliva, semen, hair roots, buccal swabs, or tissue samples.

2) DNA Extraction

Cellular material is lysed, and DNA is purified using chemical or enzymatic methods.



3) DNA Quantification

The amount and quality of DNA are measured, commonly using real-time PCR.

4) PCR Amplification

Target polymorphic regions (STR loci) are amplified using specific primers.

5) Separation and Detection

Amplified DNA fragments are separated by capillary electrophoresis.

6) Profile Analysis

Fragment sizes are analyzed to generate an individual's DNA profile.

Interpretation of DNA Profiles



Each STR locus is represented by two alleles (one from each parent). The combination of alleles across multiple loci creates a highly discriminative DNA profile. Statistical analysis is used to calculate the probability of a random match within a population.



Applications of DNA Fingerprinting

1) Forensic Science

DNA fingerprinting is widely used to:

Identify suspects in criminal investigations

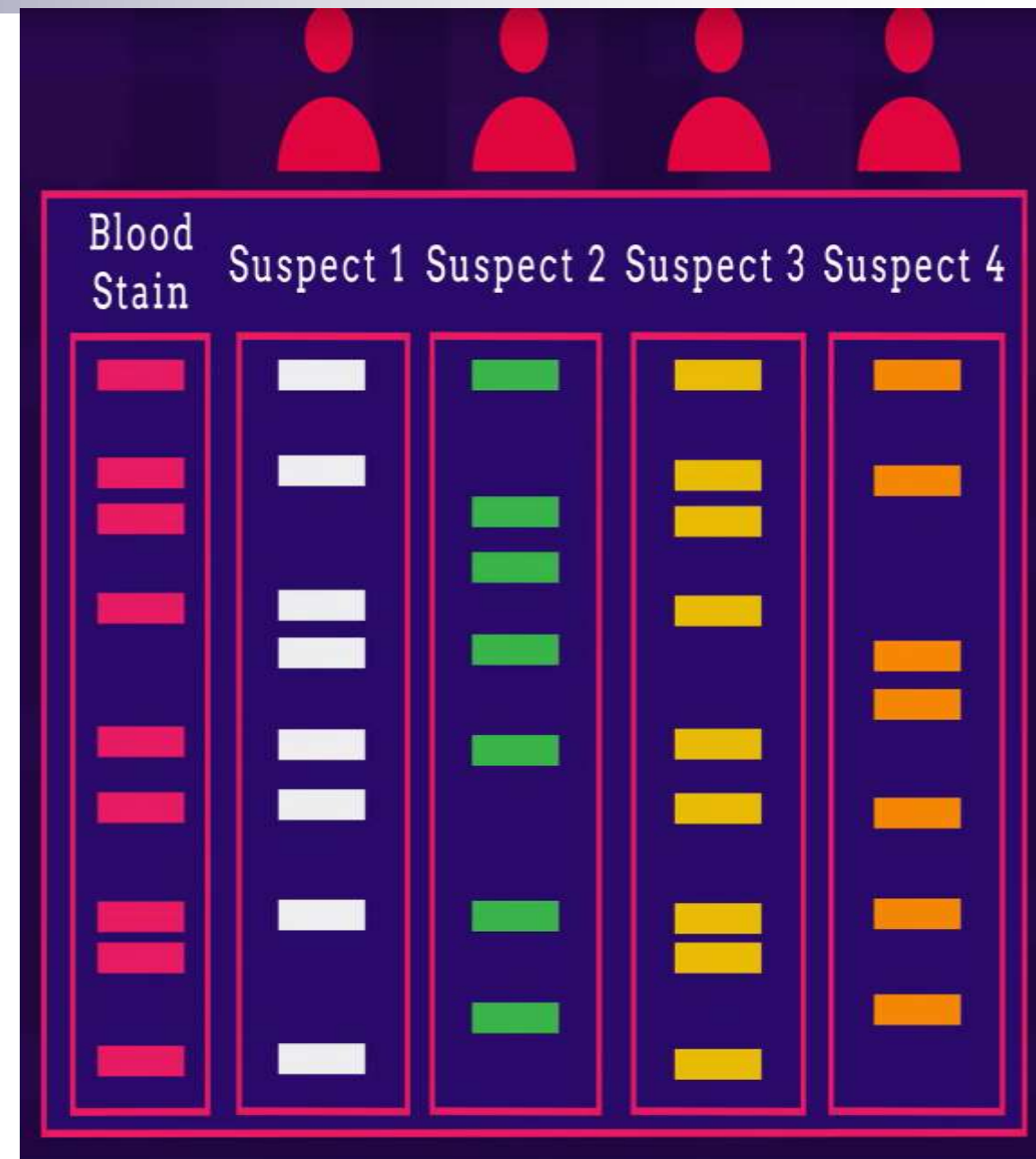
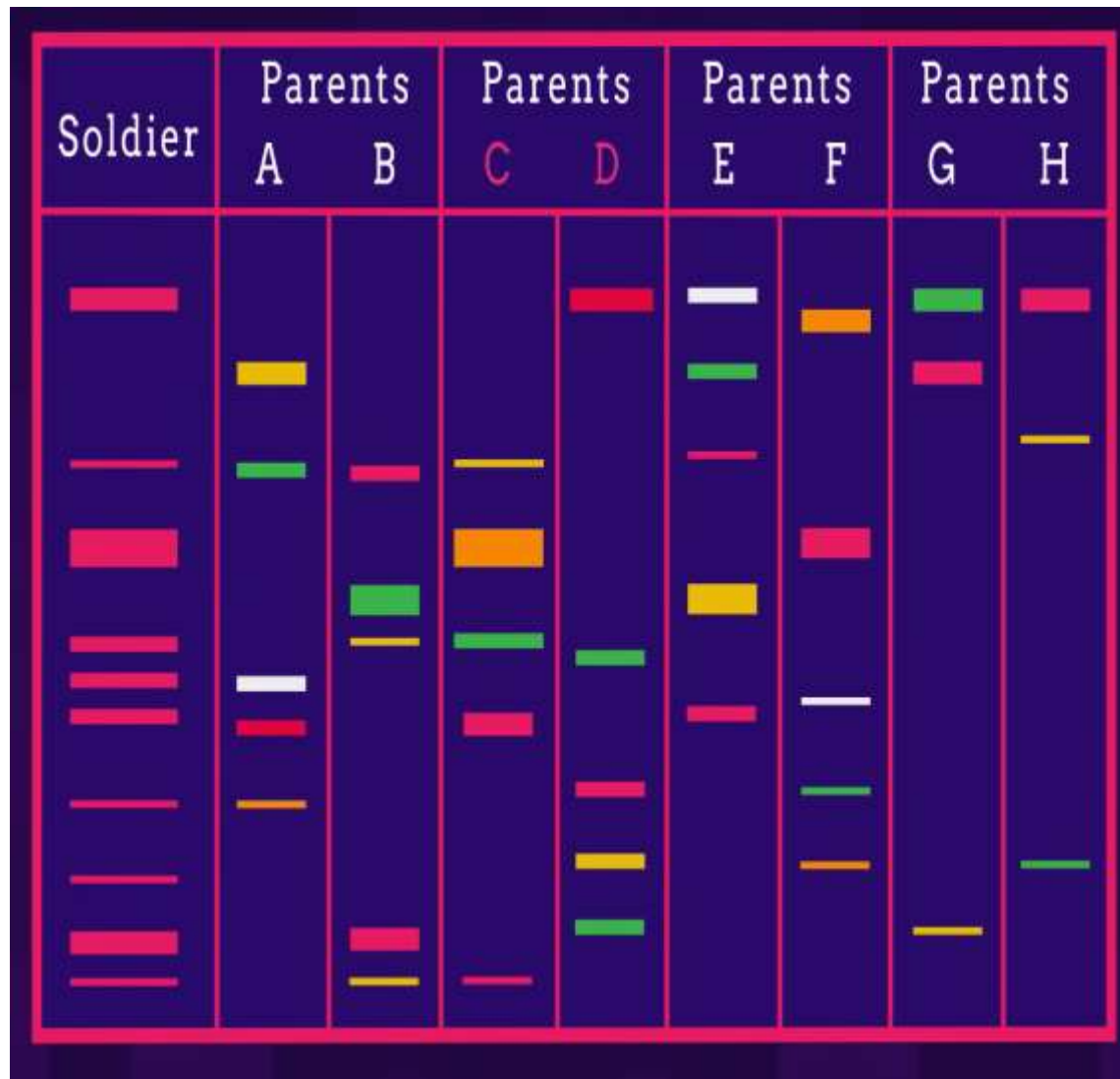
Exonerate innocent individuals

Analyze biological evidence from crime scenes

2) Paternity and Kinship Testing

DNA profiling can confirm biological relationships with high accuracy, making it valuable in paternity disputes and inheritance cases.

DNA Profiling





3) Medical and Genetic Applications

DNA fingerprinting assists in:

Identification of genetic disorders

Bone marrow transplant matching

Tracking inherited traits

4) Disaster Victim Identification

DNA analysis plays a critical role in identifying victims of natural disasters, accidents, and mass casualties.



Advantages of DNA Fingerprinting

High accuracy and specificity

Requires small amounts of DNA

Applicable to degraded samples

Provides objective and reproducible results

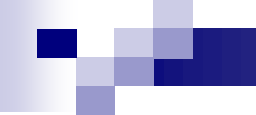
Limitations and Challenges

Risk of contamination

Ethical and privacy concerns

Requirement for specialized equipment

Interpretation complexity in mixed DNA samples



Ethical and Legal Considerations

The storage and use of DNA profiles raise ethical issues related to privacy, consent, and potential misuse. Strict regulations and ethical guidelines are essential to protect individual rights.

Recent Advances

Modern developments include:

Next-generation sequencing (NGS)–based profiling

Rapid DNA analysis systems

Improved statistical algorithms for mixture interpretation



Conclusion

DNA fingerprinting is a powerful and reliable technique for human identification. Its applications extend beyond forensic science into medicine, anthropology, and legal systems. Understanding its principles and limitations is essential for medical students and healthcare professionals, particularly in medico-legal contexts.



DNA fingerprinting was first developed by:

- A. Watson and Crick
- B. Kary Mullis
- C. Francis Collins
- D. Alec Jeffreys**
- E. James Watson

In paternity testing, a child's DNA profile must:

- A. Match only the mother
- B. Match only the alleged father
- C. Match both parents at all loci
- D. Share one allele per locus with each parent**
- E. Be identical to one parent

Thank You